

Soumya Banerjee

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Research Profile

AI researcher working across **explainable and human-centred AI, machine learning, healthcare AI, abstraction and reasoning, and computational approaches to intelligence**. Research combines machine learning with mathematical and computational modelling, with interests in trustworthy AI, human-AI collaboration, and AI systems beyond narrowly human-centric assumptions.

Academic Appointments

Senior Lecturer, University of York, UK	2026–present
Assistant Research Professor, University of Cambridge, UK	2025–2026
Senior Research Fellow & Affiliated Lecturer, University of Cambridge, UK	2022–2025
Postdoctoral Researcher, University of Cambridge, UK	2019–2022
Postdoctoral Researcher, University of Oxford, UK	2016–2018
Researcher, CSIRO, Australia	2015–2016

Earlier research positions at Harvard University/Broad Institute, Max Planck Institute, Los Alamos National Laboratory and University of New Mexico; prior industry experience as Senior Software Engineer and Technical Lead at Cognizant.

Education

PhD, Computer Science, University of New Mexico, USA, 2013
BE, Computer Science, Distinction, Nagpur University, India, 2003 – 5th rank among approximately 2,000 examinees.

Selected Research Contributions

- **Explainable and trustworthy AI:** human-interpretable machine learning and patient/public involvement frameworks for healthcare AI.
- **Healthcare AI:** machine learning on electronic healthcare records for patient stratification and mortality prediction, including work published in *NPJ Schizophrenia*.
- **Abstraction and reasoning:** research on neural networks and AI systems for abstraction and reasoning, including work presented at the NeurIPS Multimodal Algorithmic Reasoning Workshop.
- **Computational biology and complex systems:** mathematical and computational modelling of immune systems, infectious diseases, cell signalling, and socio-economic systems.

Selected Publications

- Banerjee, S. et al. “Patient and public involvement to build trust in artificial intelligence: a framework, tools and case studies.” *Patterns*, 2022.
- Banerjee, S., Lio, P., Jones, P. & Cardinal, R. “A class-contrastive human-interpretable machine learning approach to predict mortality in severe mental illness.” *NPJ Schizophrenia*, 2021.
- Bober-Irizar, M. & Banerjee, S. “Neural networks for abstraction and reasoning.” *Scientific Reports*, 2024.
- Banerjee, S. “Balancing objects and processes: Advocating pluralism in biology.” *Physics of Life Reviews*, 2025.
- Banerjee, S. & Chapman, S. J. “Influence of correlated antigen presentation on T cell negative selection in the thymus.” *Journal of the Royal Society Interface*, 2018.

Publication record: 20+ refereed journal papers plus numerous refereed conference and workshop papers.

Teaching, Supervision & Academic Leadership

Fellow of the Higher Education Academy (Advance HE). Over 10 years of higher-education teaching in machine learning, data science, reproducible research, advanced AI and data visualization. Developed and taught courses including Unsupervised Machine Learning and *Unconventional Approaches to Artificial Intelligence*.

Supervision: 17 MPhil students, 3 summer interns and co-supervision of 1 PhD student.

Academic service: ICLR Workshop HCAIR 2026 programme committee; International Conference on Artificial Immune Systems programme committee; ECML-PKDD workshop programme committee; journal and conference reviewer; PhD/MPhil/MSc recruitment panels.

Research Funding, Awards & Invited Expertise

- Co-investigator, AI@CAM pilot grants, University of Cambridge: **£150,000** (2024).
- OpenAI Researcher Access Program: **US\$3,000** API research grant (2024).
- MRC New Investigator Grant: co-investigator, under review (2025).
- 2010 Student Award for Innovation in Informatics, University of New Mexico.
- Best Poster Award, *Philosophies* International Online Conference (2025).
- Invited talks include Nokia Bell Labs, Microsoft Research, UK-India AI collaboration events, Harvard Medical School, IBM Research and CSIRO.

Technical Skills

Programming: Python, R, MATLAB, C/C++, UNIX shell, Haskell. **Data/ML:** machine learning, statistical modelling, Bayesian inference, nonlinear dynamical systems, healthcare data analysis.

Software: R/Shiny, DataSHIELD, ImageJ, CellProfiler, HPC.